IN A COMPOSITE WITH QUASI-COMPACT AND QUASI-SEPARATED OUTER MORPHISMS, THE INNER MORPHISM IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that a section of a quasi-separated morphism is quasi-compact. Invoked inside the counted proof (source lines 2056-2059). Makes the graph morphism quasi-compact.
- The base change permanence of quasi-compactness. Invoked inside the counted proof (source lines 2060-2062). Makes the projection of the fibre product quasi-compact as a base change of the outer morphism.
- The composition permanence of quasi-compactness. Invoked at the end of the counted proof (source lines 2063-2064). Combines the two quasi-compact factors.

Source reference. Original lemma lemma-base-change-quasi-compact: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1984–1984. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-base-change-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 703–703. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-composition-quasi-compact: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1994–1994. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-section-immersion: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 831–831. Navigation pointer only: the counted text refers to the Stacks item lemma-section-immersion, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Y} \to \mathcal{Z}$ be morphisms of algebraic stacks. If $g \circ f$ is quasi-compact and g is quasi-separated then f is quasi-compact.*

Proof. This is true because f equals the composition $(1, f) : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \to \mathcal{Y}$. The first map is quasi-compact by Lemma lemma-section-immersion because it is a section of the quasi-separated morphism $\mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \to \mathcal{X}$ (a base change of g, see Lemma lemma-base-change-separated). The second map is quasi-compact as it is the base change of f, see Lemma lemma-base-change-quasi-compact. And compositions of quasi-compact morphisms are quasi-compact, see Lemma lemma-composition-quasi-compact. □